GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY
FACULTY OF COMPUTING

INTAKE 41/42



Essentials of Artificial Intelligence (IT3182)

PROGRESS REVIEW

Updated project title:

HYGEIA GLOVE - AI HEALTH MONITORING SYSTEM: AI-BASED INTELLIGENT HEALTH STATUS CLASSIFICATION AND EARLY RISK PREDICTION USING VITAL SIGNS

GROUP 04

Role	**NAME**	**INDEX NUMBER**
Project Manager	R. Dineth Meditha	D/COE/25/0011
Software Developer	K.H.Prabashwari Kavindi	D/BIT/24/0063
Data Analyst	H.K.K.Aberathna	D/BIS/24/0026
Quality Assurance Engineer	T.S Karunrathne	7165

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Plagiarism Declaration

We recognize that plagiarism is defined as taking someone else's ideas, work or information without giving it proper credit. We are aware of the significance of proper referencing, citing and ethical academic writing. We confirm we have followed the referencing guidelines and that our work will not be copied or presented by others as theirs.

NAME	INDEX NUMBER	SIGNATURE
R. Dineth Meditha	D/COE/25/0011	
K.H.Prabashwari Kavindi	D/BIT/24/0063	
H.K.K.Aberathna	D/BIS/24/0026	
T.S Karunrathne	7165	

1. Updated Problem Statement

Monitoring of vital signs is essential in assessing any change in a person's physiological state. Traditional systems of monitoring usually rely on manual observations or thresholds of alarms. Even though these approaches are helpful in identifying clearly abnormal values, they fail to consider several indicators or show the progression of the health status.

This problem can be solved through the development of the HYGEIA GLOVE system – a smart glove using ESP32 board that monitors the user's body temperature, SpO_2, and pulse rate values. The values collected via the glove are transferred through Wi-Fi to a Flask server where the values are classified into Normal, Warning, or Critical according to an AI model.

In addition to that, the project features a dashboard showing the current vital signs and 30 minutes trend. The objective of the system is to act as an intelligent health monitoring and decision-making prototype and not substitute medical professionals' work.

2. Changes Made After the Proposal

During the process of implementation, there were made some changes to the initial proposal in order to build a working prototype.

Initial Proposal	Current Implementation
Inputs (heart rate, SpO_2 and temperature)	MLX90614 and MAX30102 sensors
Supervised ML classification	Bidirectional LSTM neural network implementation with Conv1D layer
Historical trend analysis	Historical data of 30 min is collected
Fuzzy logic risk assessment	Not yet implemented
Frontend using React.js	HTML/CSS/JavaScript dashboard running in the Flask server
Database is needed	For now, the in-memory rolling buffer is used as storage
Option to use ESP32	ESP32 Smart Glove is used
General health monitoring prototype	Real-time sensor → AI → dashboard pipeline developed

3. The Main AI Techniques

At the moment, the main technique that is being implemented is deep learning classification.

3.1 Supervised Health Classification

This solution considers the following vital signs:

- Body Temperature
- SpO_2
- Pulse Rate

Besides the vital signs mentioned above, there are three additional engineered features:

- Temperature/SpO_2 ratio
- Pulse/SpO_2 ratio
- Health index

Thus, the model gets 6 features as input.

The final output is the class of health status of the patient:

Normal → Warning → Critical

3.2 Feature Engineering

Feature engineering is used to add more information to the model in terms of dependencies between vital signs, not separate sensor measurements.

3.3 Data Scaling

Robust Scaler is used to scale the input features before giving them to neural network.

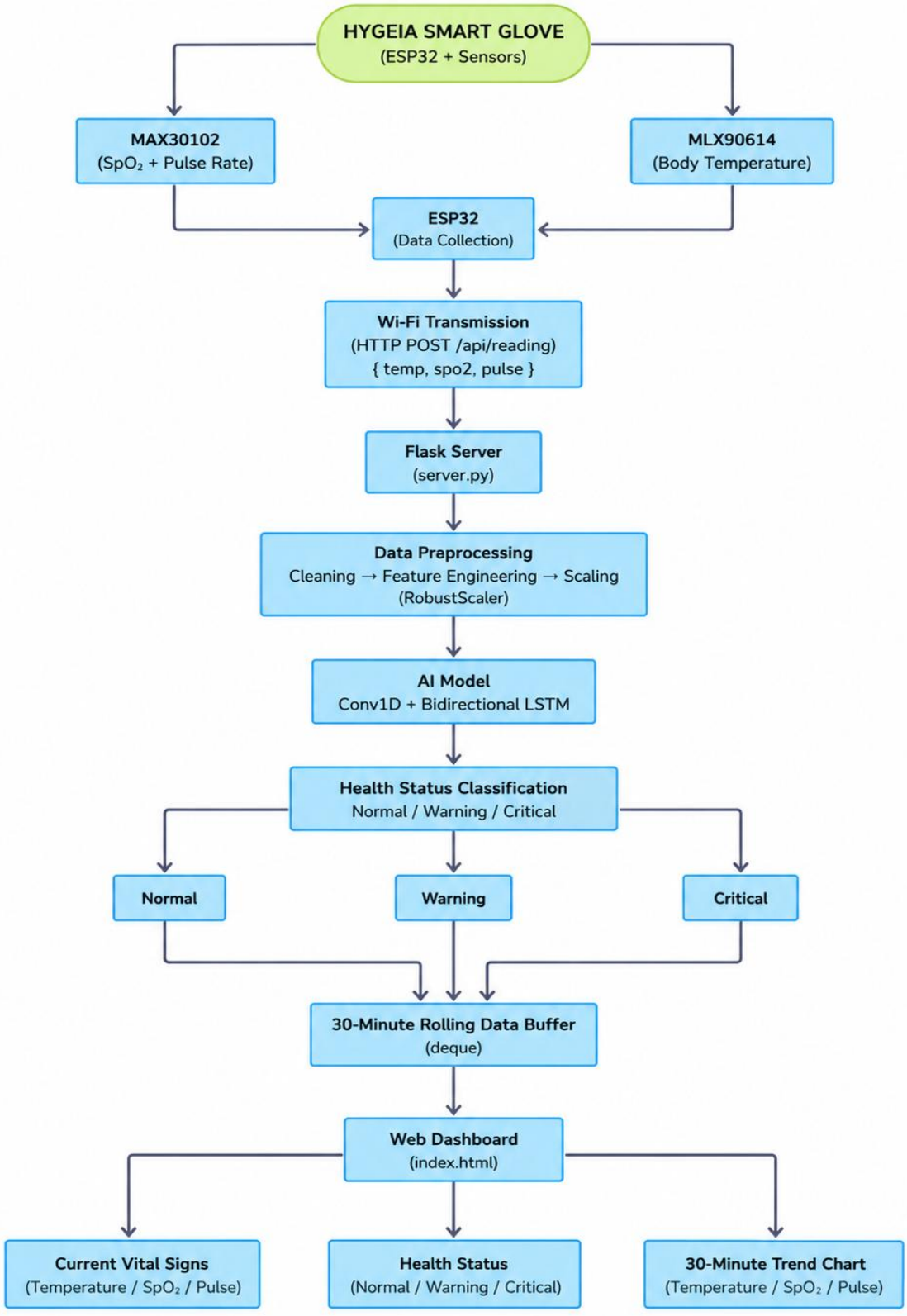
3.4 Temporal Monitoring

Most recent sensor readings are stored in a rolling buffer and visualized through dashboard. Thus, user can see how temperature, SpO_2 and pulse change in time.

3.5 Fuzzy Logic and Early-Risk Assessment

Fuzzy logic is included in the proposal to combine physiological signs in an aggregated risk assessment and do temporal analysis in order to assess Stable, Increasing or Decreasing risk.

4. System Architecture / Workflow



5. Dataset Details

The following files were used for the project:

data/glove_readings.csv

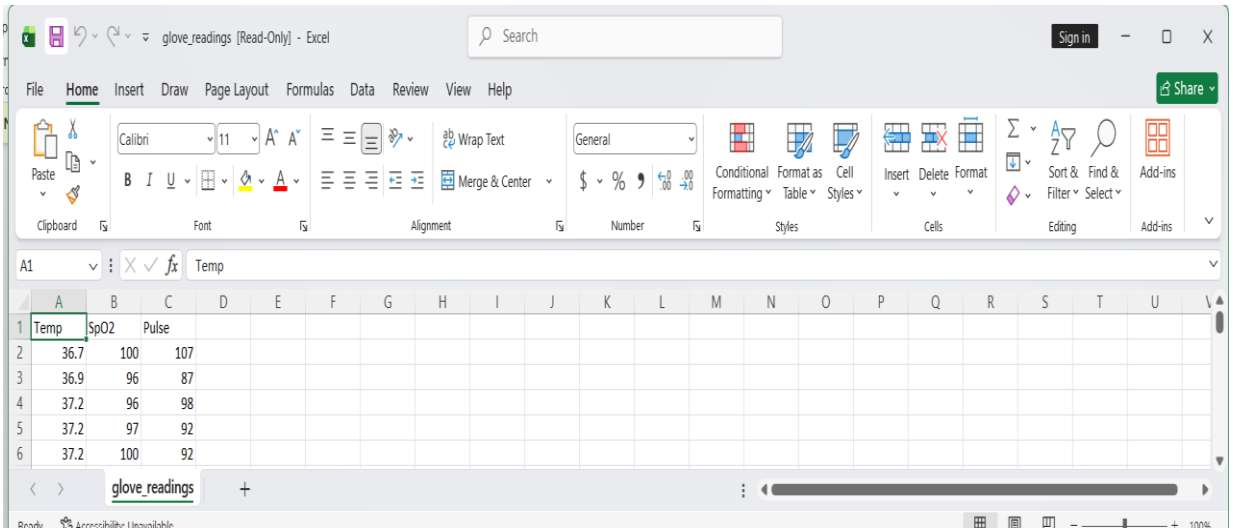
The important raw features are as follows:

Feature	Description
Temp	Body temperature
SpO2	Blood oxygen saturation
Pulse	Pulse rate (BPM)

Additionally, three engineered features were constructed:

Engineered feature	Purpose
Temp_SpO2_ratio	Ratio of temperature to SpO2
Pulse_SpO2_ratio	Ratio of pulse to SpO2
Health_Index	Index reflecting the state of the person based on vital signs

The dataset was prepared and processed prior to training. The prepared features were scaled with the RobustScaler and fed into the neural network.
The current drawback is the lack of extreme abnormal observations in the dataset.



6. Health Classification Rules

The below criteria can be used in categorizing health status.

Status	Criteria
Normal	SpO2 >=95%, Body Temperature 36.0 – 38.5, Pulse 55-120 BPM
Warning	SpO2 90-94%, Body Temperature slightly out of range, Pulse out of normal range
Critical	SpO2<90%, Body Temperature above 40 degrees or less than 35, Pulse greater than 150 or less than 40

This will help in dataset labeling and is also an alternative classification

7. Neural Network Structure

The current architecture makes use of the combination of Conv1D (Convolutional Neural Network) layers and BiLSTM (Bidirectional Long Short-Term Memory) layers.

Layer	Configuration
Input	Temp, SpO_2, Pulse + other engineered features
Conv1D	64 filters
Batch Normalization	Applied
Dropout	0.3
Conv1D	128 filters
Batch Normalization	Applied
Dropout	0.3
BiLSTM	64 units
BiLSTM	32 units
Dense	128 units
Dense	64 units
Output	Softmax of 3 classes

The final Softmax layer predicts the probability for Normal / Warning / Critical

8. Partial Implementation

A considerable part of the prototype has already been developed.

Implemented Components

ESP32 Smart Glove

- ESP32 firmware developed

- MAX30102 sensor integrated
- MLX90614 sensor integrated
- Wi-Fi communications realized
- Sensor data sent to Flask by HTTP POST

AI Component

- Dataset preprocessing developed
- Feature engineering developed
- RobustScaler developed
- Conv1D + BiLSTM model developed
- Model trained
- Model saved as hygeia_ai_model_final.h5
- Scaler saved as hygeia_scaler.pkl
- Model tested

Backend

- Flask server developed
- REST API developed
- Live sensor data supported
- Demo-data support developed
- Rolling data buffer developed

Frontend

- Live web dashboard developed
- Vital signs shown
- Health classification shown
- Trend visualization developed

API Endpoints Developed

Method	Endpoint	Description
GET	/	Opens dashboard
POST	/api/reading	Receive ESP32 readings
GET	/api/data	Recent readings retrieval
GET	/api/current	Latest readings retrieval
GET	/api/status	Server status check
POST	/api/sample/start	Starting of simulation data
POST	/api/sample/stop	Stopping of simulation data
POST	/api/clear	Deletes current buffer



9. Current AI Model Results

The present model performed in:
Overall Accuracy: 90.91%
Average Prediction Confidence: about 93.5%

The classification performance is as follows:

Class	**Precision**	**Recall**	**F1 Score**	**Samples**
Normal	0.9322	0.9483	0.9402	58
Warning	0.8333	0.7895	0.8108	19
Critical	0.0000	0.0000	0.0000	0

An important issue to be noted is that there are no Critical cases in the present test

10. Screenshots / Sample Outputs

Figure 1 – AI Model Training Results

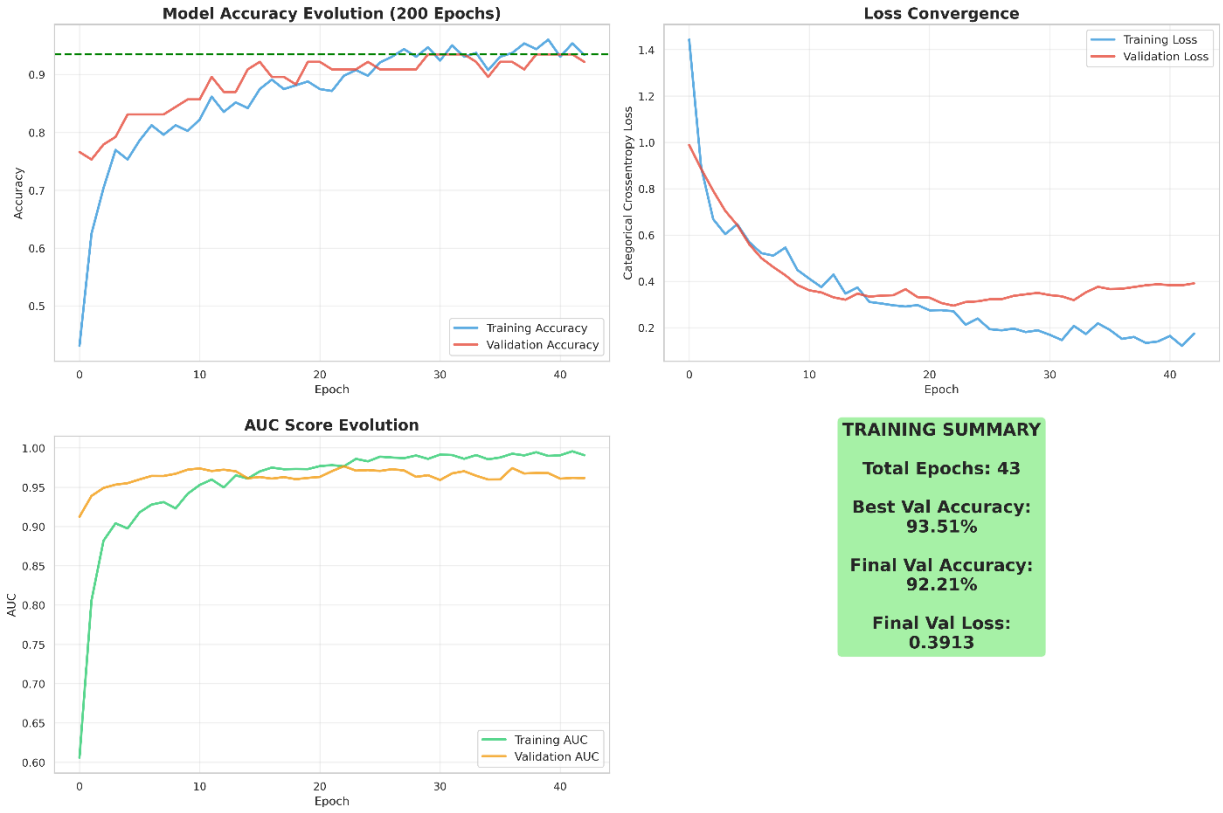


Figure 2 – Model Evaluation

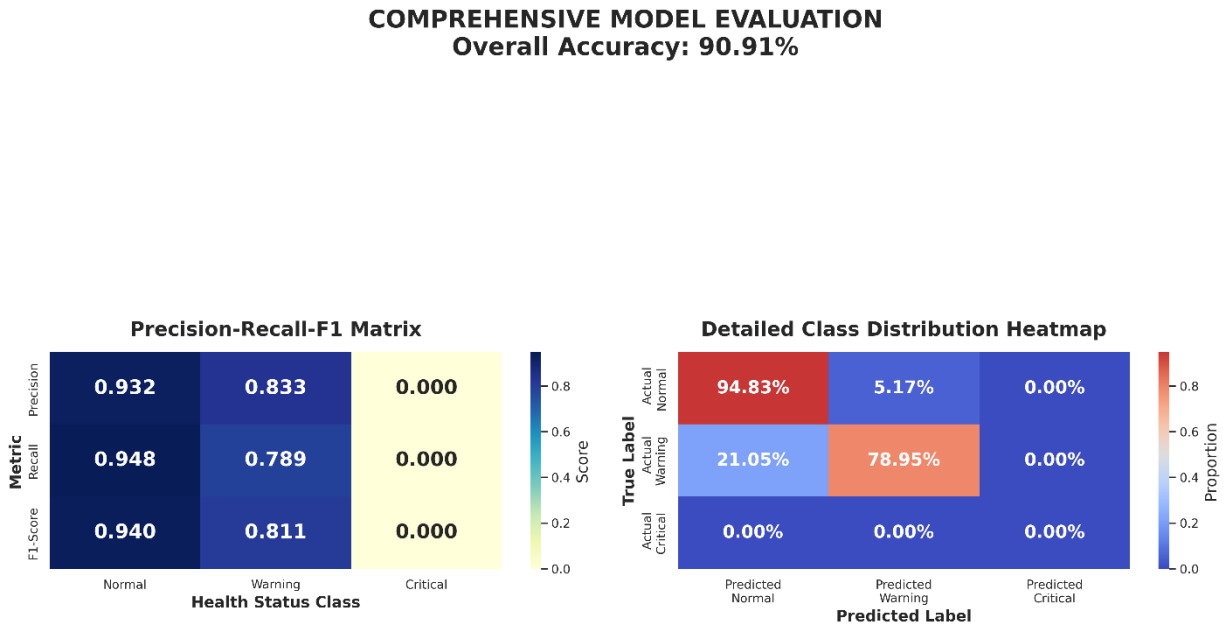
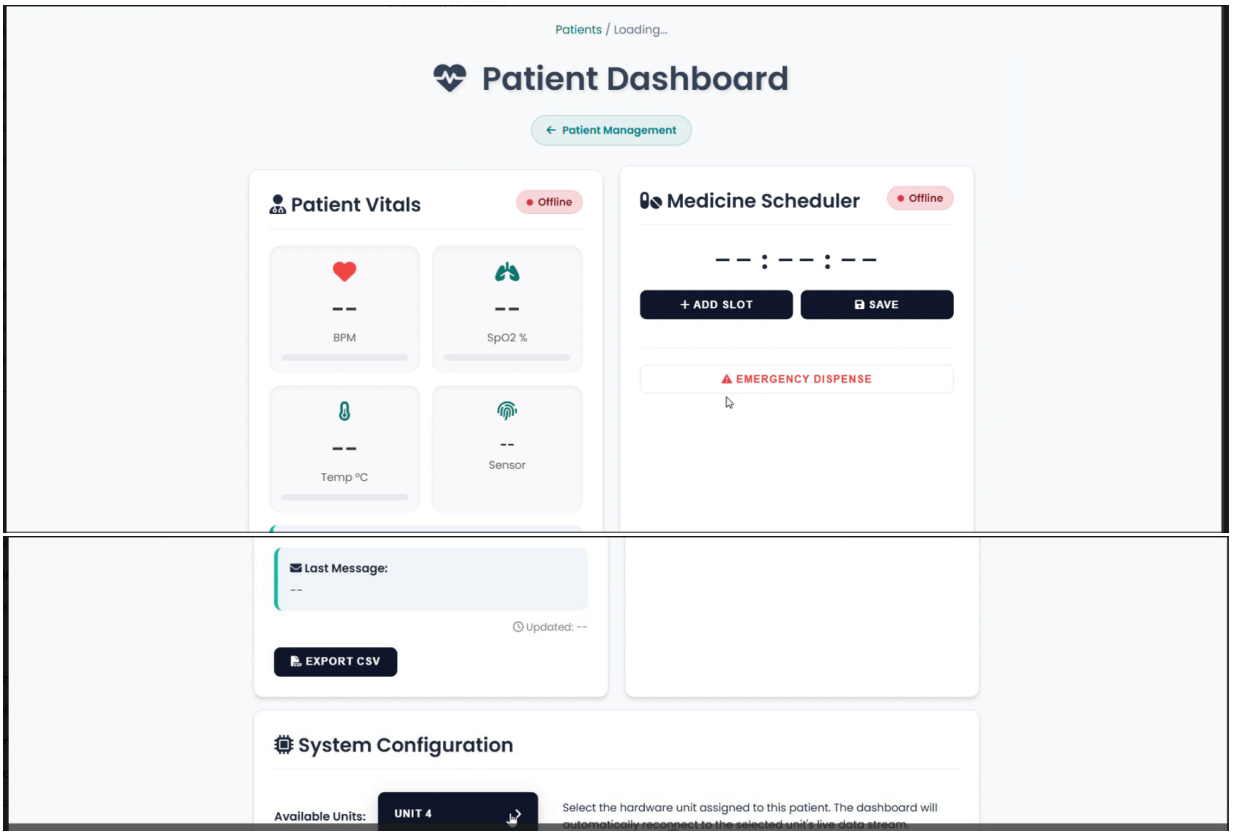


Figure 3 – HYGEIA Web Dashboard



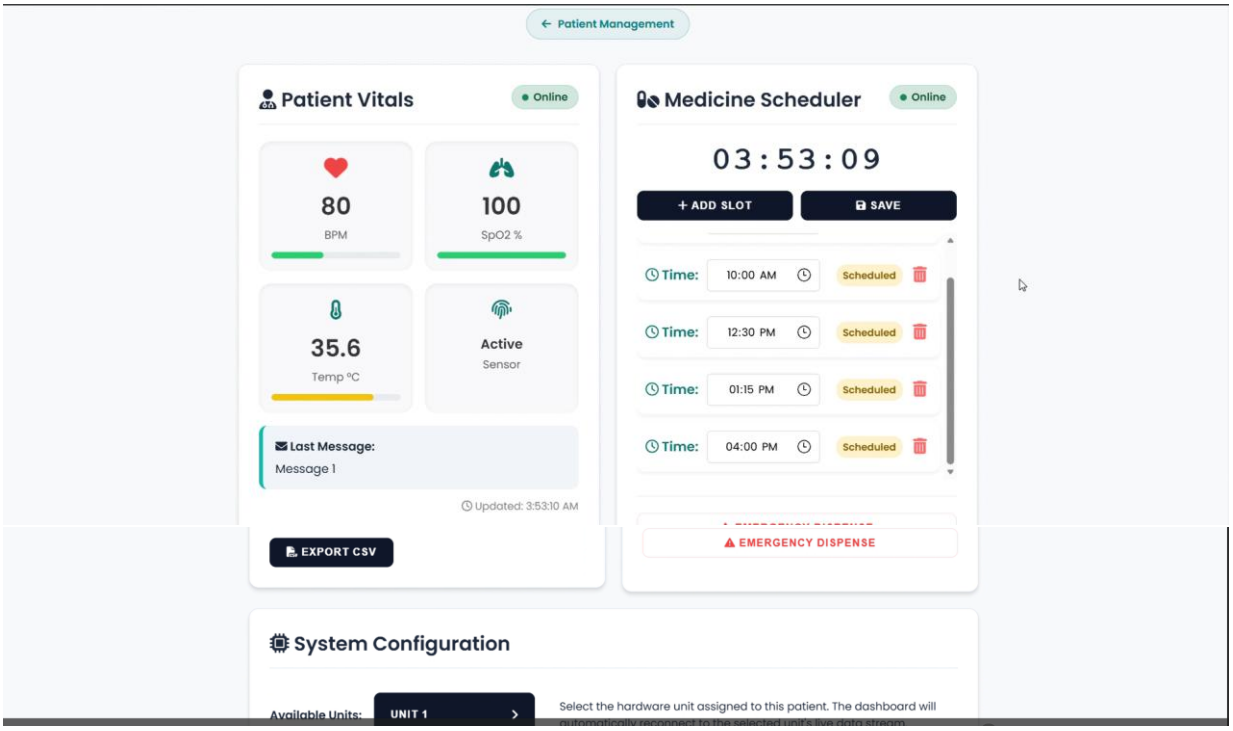


Figure 4 – 30-Minute Vital-Sign Trend

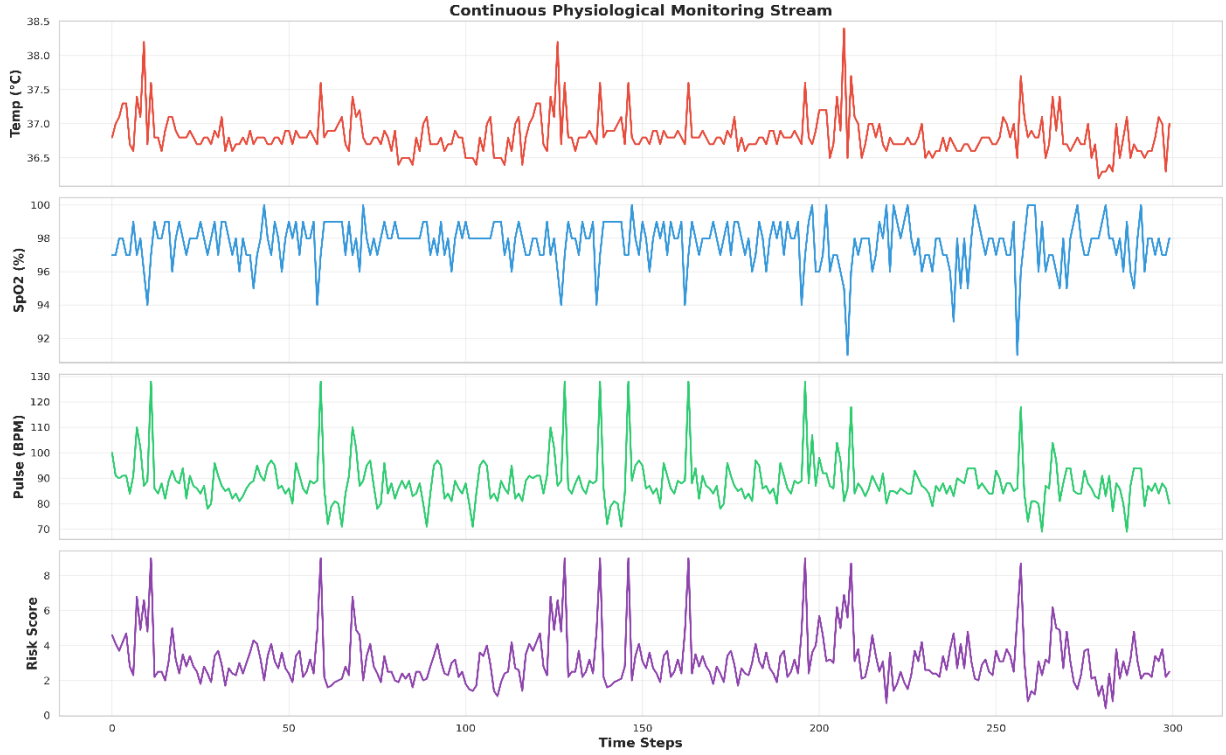


Figure 5 – ESP32 Smart Glove





11. Problems Faced During Development

Some of the issues that arose during the process include the following:

- The sensor readings might depend on finger movement and sensor noise.
- The temperature sensors might need calibration.
- There needs to be a network between the ESP32 and Flask servers.
- The IP address of the computer needs to be set up in the ESP32's firmware.
- The dataset size is relatively small.
- There are not enough Critical-class samples for proper three-class evaluation.
- The real-time communication will depend on a stable Wi-Fi connection.
- The risk component using fuzzy logic needs further integration.
- Explicit early risk prediction as Stable/Increasing/Decreasing has to be done.
- The current rolling data need to be saved in the database.

This list of issues corresponds with the warning in the proposal about the impact of dataset size, synthetic data, sensor noise, false positives, and false negatives on the prototype.

12. Remaining Work Before Final Submission

Prior to the final delivery, the following will be done:

1. Conduct the fuzzy-logic risk assessment.
2. Define the fuzzy variables, membership functions and rules.
3. Complete explicit early-risk and trend prediction.
4. Produce outputs that can be classified as Stable, Increasing or Decreasing.
5. Improve the dataset especially on the Critical class.
6. Retrain and evaluate the model if necessary.
7. Conduct end-to-end testing with the real ESP32 glove.
8. Test normal, warning, and critical scenarios.
9. Complete the integration of the dashboard for the remaining risk information.
10. Conduct final tests on the system and user interface.
11. Document the ethical, privacy, reliability and AI limitations.
12. Complete the final report, slides and video.

13. Updated Contribution of Each Member

Member	**Main Area**	**Contribution to Current Work**
R. Dineth Meditha	ML Classification & AI Model Development	Dataset preprocessing, feature engineering, AI model development, training and evaluation
K.H. Prabashwari Kavindi	AI Integration, Frontend & Dashboard	AI prediction integration, dashboards, graphs and alerting
H.K.K. Aberathna	Backend & Integration	Flask REST API, real-time data processing and integration
T.S. Karunrathne	Risk Prediction & Fuzzy Logic	Risk trend analysis & fuzzy-logic implementation

14. Conclusion

The HYGEIA GLOVE – AI Health Monitoring System prototype allows now collecting data on temperature, SpO2, and pulse rate via ESP32 sensors, sending them to the Flask server, health classification using AI, REST API communication, 30 minutes' rolling history, and live web dashboard. Currently, the accuracy of the Conv1D + Bidirectional LSTM model is at 90.91% with the approximate 93.5% of mean prediction confidence. But more development is needed because the evaluation at this moment lacks Critical class samples, and fuzzy-logic risk assessment as well as early-risk trend classification are not finished yet. Future work will focus on finishing AI components, increasing the coverage of the dataset, performing hardware evaluation, evaluating all three classes of health status, and completing the final prototype.